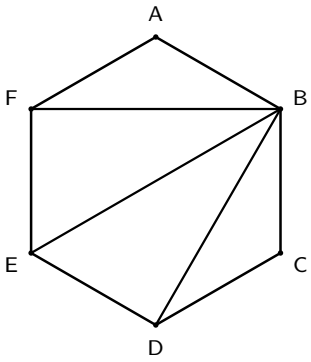
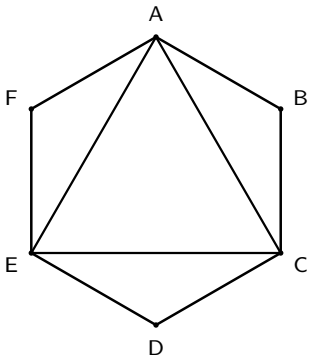
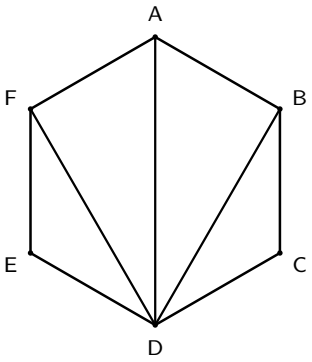
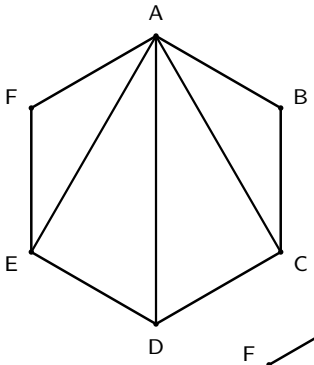
Fill the whole polygon with triangles, using straight lines between printed corners. Lines inside a polygon may meet at corners but may not cross. Keep the corner labels fixed, even if you turn the page. Use tracing paper to change printed lines.

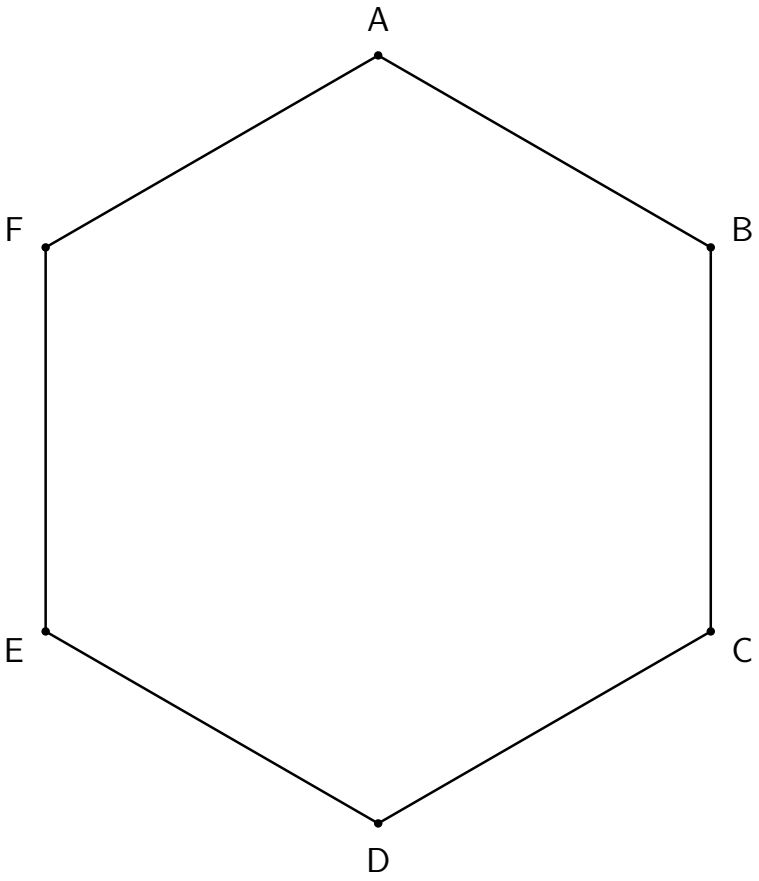
**Problem 4:** Change each of the three starting fillings into the fourth filling using as few flips as possible. Explain why no shorter routes can work.

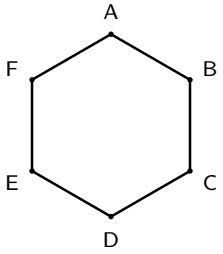


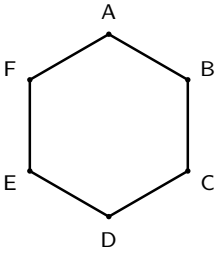



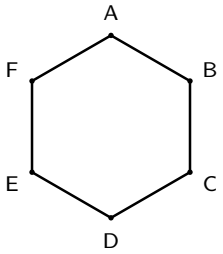



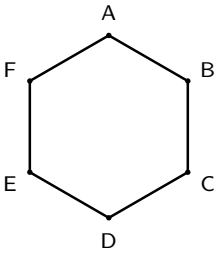



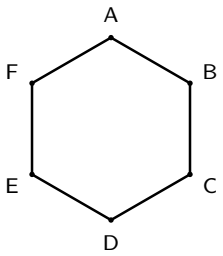



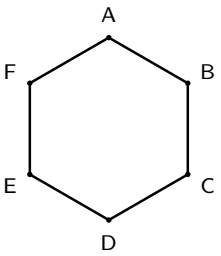



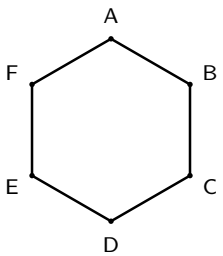



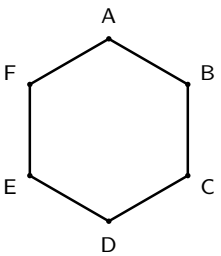



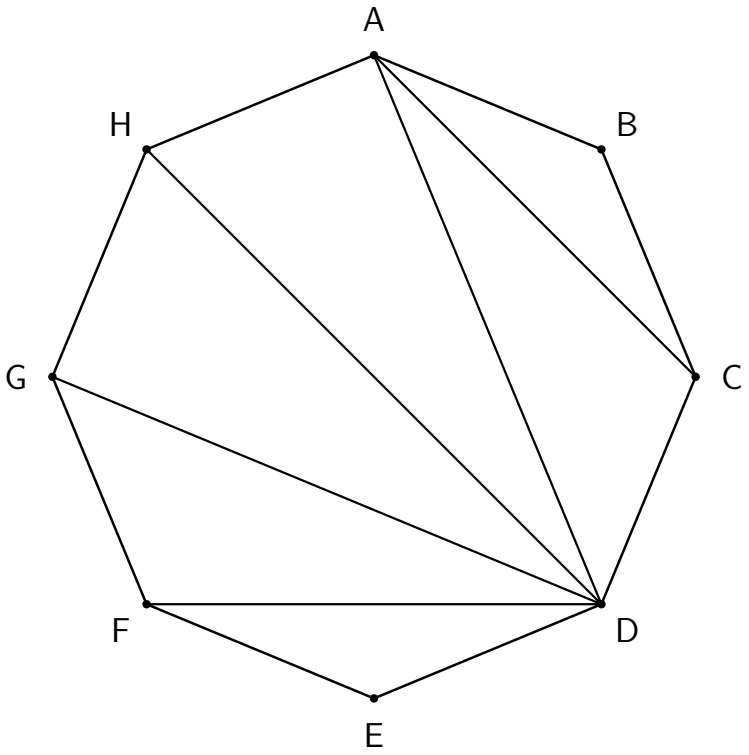



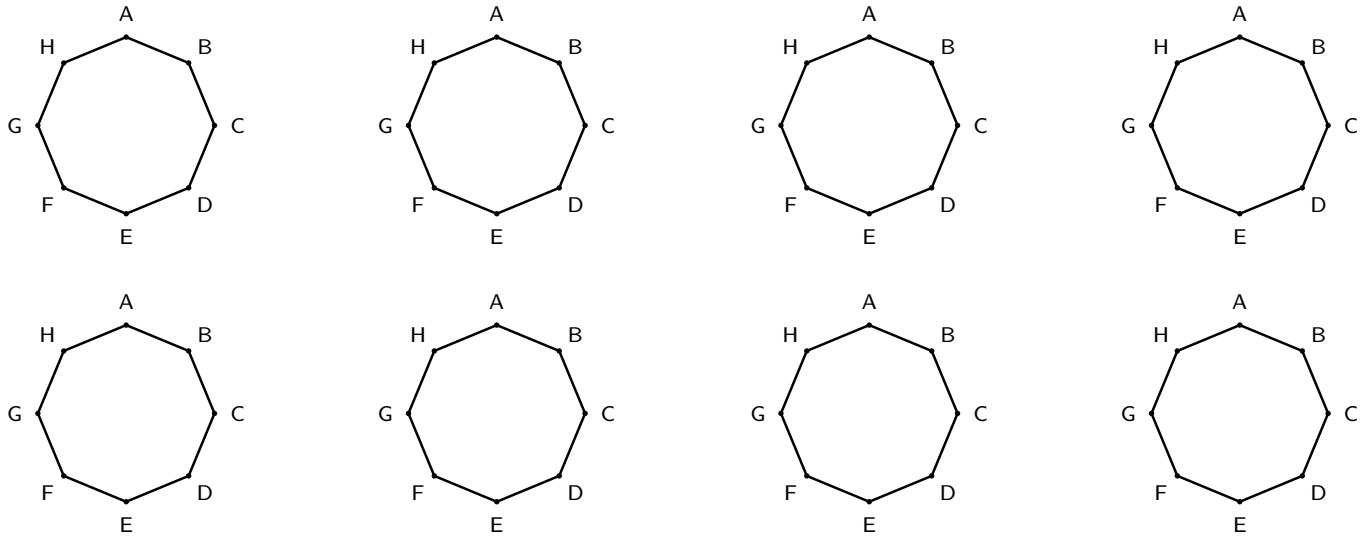



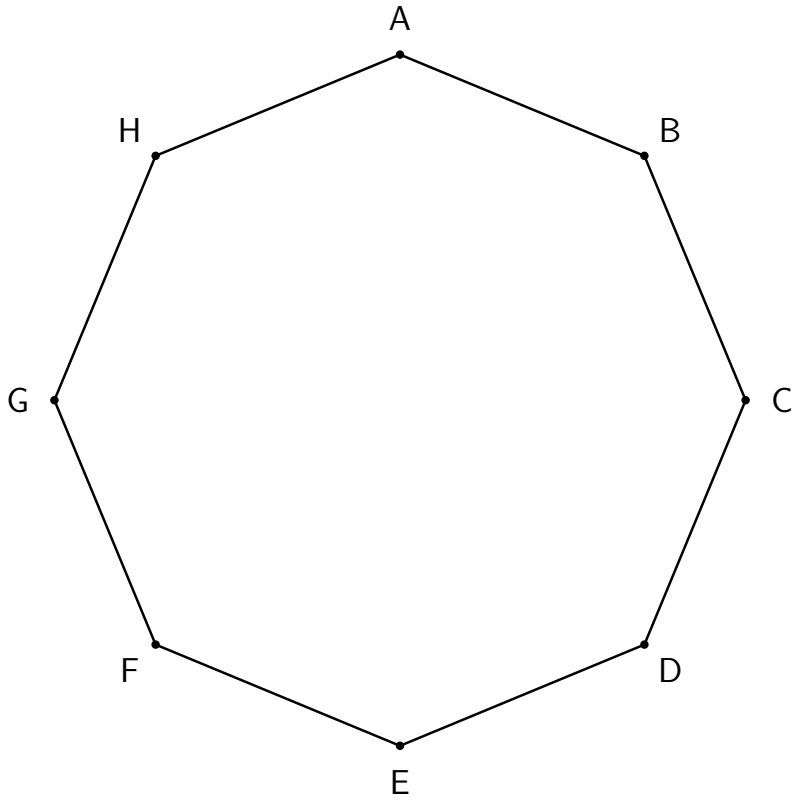



**Problem 5:** A filling whose inside lines all meet at one corner is called a fan. Find the fewest flips from this filling to a fan at A, and to a fan at E. Explain why each route is shortest.



**Problem 6:** Find a rule for the fewest flips needed to turn any filling into a fan at A. Explain why your rule works for every starting filling and every number of corners, when all corners point outward.



**Problem 7:** Find a route of flips between these two fillings. Can any two fillings of the same polygon be joined by flips? Explain why your answer works for any polygon with no inward-pointing corners.

